PROJECT REPORT

On

Smart Voice and Gesture Controlled Wheelchair With IoT Monitoring And Obstacle Detection System

A project report submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF TECHNOLOGY

In

ELECTRONICS AND COMMUNICATION ENGINEERING

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2022-2026

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This is hereby declared that this report is being submitted in fulfilment of the "**SMART VOICE AND GESTURE CONTROLLED WHEELCHAIR WITH IoT MONITORING AND OBSTACLE DETECTION SYSTEM"** project and carried out by us under the guidance of **Mr.T.DURGA RAO, M. Tech., (Ph. D**), Assistant professor (Adhoc) **.**We further declare that to our knowledge the structure and content of this report are original and have not been submitted before any purpose.

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ACKNOWLEDGEMENT

It is our proud privilege and duty to express heartfelt gratitude to supervisor **Mr.T.DURGA RAO, M.Tech., (Ph. D),** for his continous support, valuable guidance and encouragement throughtout the course of this project propect titled "**Smart Voice and Gesture Controlled Wheelchair with Obstacle Detection**". His expertise and insights were instrumental in the sucessful completion of this work.

Deep sense of appreciation goes to **Dr.K.Aparna, M.Tech.,Ph.D.,** Associate Professor and Head of the Department of E.C.E JNTUA College of Engineering, Kalikiri, for her constant encouragement and insightful suggestions throughout the project period.

We also thankful to **Prof.M.Venkateswara Rao**, Prinicipal of JNTUACEK, for providing all the necessary facilities and a conductive environment that helped us carry out the project work effectively and within the stipulated time.

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ABSTRACT

This project presents a **Smart Voice And Gesture Controlled Wheelchair with IoT Monitoring And Obstacle Detection System**. The system is designed to assist physically challenged and elderly people by providing an intelligent and safe wheelchair control mechanism. The wheelchair can be controlled using both **gesture and offline voice commands**, making it easy to operate without physical effort. The **ESP32 microcontroller** is used as the main controller and runs multiple tasks using **FreeRTOS** for efficient and real-time operation.

The system uses an **MPU6050 sensor** for gesture detection and a **VC02 offline voice recognition module** for voice control. Two **ultrasonic sensors** are placed at the front and back of the wheelchair to detect obstacles and prevent collisions. If an obstacle is detected within **20 cm**, the system automatically stops the motors and activates a **buzzer alert** to warn the user, even if the wheelchair is not moving. This ensures continuous safety monitoring.

The wheelchair movement and obstacle distances are monitored using the **Blynk IoT dashboard**, where **V0 displays wheelchair status (Forward, Backward, Left, Right, Stopped)**, **V1 shows front obstacle distance**, and **V2 shows back obstacle distance** in real time. The system also prioritizes **voice commands over gesture control** and includes an **emergency stop feature** for immediate motor shutdown and safety notification.

The proposed system provides a **low-cost, efficient, and user-friendly smart wheelchair** with real-time monitoring and enhanced safety features. This solution improves mobility assistance and ensures safe navigation using modern embedded and IoT technologies.

The proposed system is a smart wheelchair controlled using **gesture and voice commands** with **real-time obstacle detection and IoT monitoring using Blynk**. The system is designed using **ESP32 microcontroller with FreeRTOS**, which allows multiple tasks like voice control, gesture control, obstacle detection, and Blynk communication to run simultaneously and efficiently.

The wheelchair can move **forward, backward, left, right, and stop** based on voice or gesture commands. The system continuously monitors obstacles using **two ultrasonic sensors (front and back)** and ensures safety by automatically stopping the motors and activating the buzzer when an obstacle is detected within **20 cm distance**. The **Blynk IoT dashboard** displays the wheelchair status and obstacle distances in real time.

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CHAPTER 1: INTRODUCTION

In recent years, advancements in embedded systems, Internet of Things (IoT), and intelligent control technologies have significantly transformed assistive devices designed for physically challenged individuals. Among these devices, wheelchairs play a vital role in enhancing mobility and independence. However, traditional wheelchairs, whether manual or basic electric models, often require considerable physical effort or continuous manual control, which may not be feasible for individuals with severe disabilities.

To address these limitations, smart wheelchairs have emerged as an innovative solution that integrates modern technologies such as sensors, microcontrollers, wireless communication, and intelligent control systems. These wheelchairs are designed to provide hands-free operation, improved safety, and enhanced user convenience. By incorporating features such as voice recognition, gesture control, and obstacle detection, smart wheelchairs enable users to navigate their environment more efficiently and independently.

The proposed system focuses on developing an IoT-enabled smart wheelchair that allows users to control movement through voice commands and hand gestures. The system utilizes a microcontroller as the central processing unit, which interprets inputs from various sensors and executes appropriate motor control actions. Additionally, ultrasonic or proximity sensors are integrated to detect obstacles in real time, ensuring user safety by preventing collisions.

The integration of IoT technology further enhances the functionality of the system by enabling remote monitoring, data transmission, and emergency alert features. For instance, caregivers or family members can track the wheelchair's status or receive alerts in case of abnormal conditions. This makes the system not only a mobility aid but also a safety-enhancing and assistive communication device.

Overall, the proposed smart wheelchair aims to provide a cost-effective, user-friendly, and efficient mobility solution that improves the quality of life for physically disabled individuals. By leveraging modern technological advancements, the system seeks to reduce dependency on caregivers and promote greater independence and confidence among users.

1.1 Problem Statement

Despite the availability of conventional and electric wheelchairs, several challenges still persist that limit their effectiveness and usability for individuals with severe physical disabilities.

Firstly, manual wheelchairs require significant physical effort to operate, which makes them unsuitable for users with limited upper body strength or mobility impairments. Even electric wheelchairs, although easier to use, typically rely on joystick-based controls, which may not be accessible to individuals with hand movement limitations.

Secondly, most existing wheelchairs lack intelligent control mechanisms such as voice or gesture-based operation. This restricts their usability for individuals who cannot perform precise manual control actions. As a result, many users remain dependent on caregivers for mobility, reducing their independence and quality of life.

Another major issue is the lack of safety features in traditional systems. Wheelchairs without obstacle detection mechanisms are prone to collisions, especially in crowded or unfamiliar environments. This can lead to accidents, injuries, and reduced confidence among users while navigating.

Furthermore, existing solutions often do not provide real-time monitoring or communication capabilities. In emergency situations, there is no efficient way to alert caregivers or track the user's condition and location. This lack of connectivity poses a significant risk, particularly for elderly or severely disabled individuals.

In addition, many advanced smart wheelchair systems available in the market are expensive and complex, making them inaccessible to a large portion of users, especially in developing regions.

Therefore, there is a need for a cost-effective, intelligent, and user-friendly smart wheelchair system that:

- Enables hands-free control using voice and gesture recognition
- Incorporates obstacle detection for enhanced safety
- Provides IoT-based monitoring and alert mechanisms
- Reduces dependency on caregivers

- Is affordable and easy to use

The proposed system aims to address these challenges by developing an integrated smart wheelchair that combines accessibility, safety, and connectivity into a single efficient solution.

1.2 Objectives of the Project

The main objectives of the Smart Voice and Gesture Controlled Wheelchair with Obstacle Detection are:

- To design and develop a smart wheelchair that can be controlled using voice commands for hands-free operation
- To implement a gesture-based control system that allows users to navigate the wheelchair using simple hand movements
- To integrate obstacle detection sensor (such as ultrasonic) to ensure safe movement by detecting and avoiding obstacles in real time
- To enhance user independence and mobility, especially for individuals with severe physical disabilities or limited motor control
- To develop a reliable and efficient control system using a microcontroller for processing inputs and controlling motor actions
- To ensure the system is cost-effective and user-friendly, making it accessible to a wider range of users
- To improve overall safety, comfort, and ease of use compared to conventional wheelchairs
- To reduce dependency on caregivers by providing an intelligent and automated mobility solution
- To integrate a real-time emergency alert system that sends notifications to a predefined caregiver or family member in critical situations.
- To provide voice-controlled auxiliary features such as lighting control to improve usability in different environments

CHAPTER 2: LITERATURE SURVEY

The development of smart wheelchairs has gained significant attention in recent years due to the increasing need for assistive mobility solutions for physically disabled and elderly individuals. Researchers have explored various technologies such as voice recognition, gesture control, obstacle detection, and IoT integration to enhance wheelchair functionality, safety, and user independence.

A comprehensive review conducted between 2017 and 2025 analyzed multiple smart wheelchair systems focusing on control mechanisms and obstacle detection techniques. The study highlighted that voice control systems provide hands-free operation but often face challenges such as reduced accuracy in noisy environments and limited support for different languages. Joystick-based systems, although reliable, are not suitable for users with severe motor impairments. Obstacle detection technologies, including ultrasonic sensors and LiDAR, demonstrated good accuracy in controlled environments but lacked standardized performance evaluation in real-world scenarios. The study also emphasized the need for user-centered design and cost-effective solutions for wider adoption.

Early research on voice-controlled wheelchairs primarily focused on integrating speech recognition modules with microcontrollers to enable basic directional movement. These systems improved accessibility for users with limited hand movement but were constrained by low processing capabilities and dependence on predefined voice commands. Additionally, environmental noise significantly affected system performance, limiting practical usability.

Several studies have explored Arduino-based smart wheelchair systems that combine voice control with obstacle detection mechanisms. These systems typically utilize ultrasonic sensors to detect obstacles and prevent collisions. While such designs are cost-effective and simple to implement, they often lack advanced features such as real-time monitoring, adaptive control, and intelligent decision- making capabilities.

Recent advancements have introduced multi-modal control systems that combine voice, gesture, and touch-based inputs. For example, smartphone-based control systems allow users to operate the wheelchair using voice commands, hand gestures, or mobile applications via Bluetooth communication. These systems enhance flexibility and user interaction but depend heavily on external devices and stable connectivity, which may not always be reliable.

Gesture-controlled wheelchairs have also been widely studied, where accelerometers or motion sensors are used to interpret hand movements into navigation commands. These systems provide intuitive control for users with speech impairments. However, they may suffer from accuracy issues due to unintended hand movements or sensor noise. Some systems also incorporate fall detection and remote-control features to improve safety and usability.

In recent developments, researchers have integrated Artificial Intelligence (AI) and IoT technologies into smart wheelchairs to enhance performance and functionality. AI-based systems use computer vision and deep learning algorithms for advanced obstacle detection and navigation. IoT integration enables real-time monitoring of user health parameters and sends alerts during emergencies. These systems significantly improve safety and autonomy but increase system complexity and cost.

Additionally, hybrid systems combining voice control, gesture recognition, and obstacle detection with renewable energy sources such as solar power have been proposed. These systems aim to provide sustainable and efficient solutions while improving user convenience and reducing operational costs.

Despite these advancements, several limitations still exist in current smart wheelchair systems. Most systems are tested in controlled environments and may not perform reliably in real-world conditions. High costs, technical complexity, and lack of adaptability to different user needs hinder widespread adoption. Furthermore, limited focus on user experience, affordability, and maintenance remains a significant challenge.

CHAPTER 3: EXISTING METHOD

3.1 Introduction

Mobility assistance is a fundamental requirement for individuals with physical disabilities, elderly people, and patients recovering from injuries or neurological disorders. Wheelchairs serve as one of the most widely used assistive devices, enabling users to perform daily activities with a certain level of independence. Over time, the design and functionality of wheelchairs have evolved from simple mechanical structures to electrically powered systems aimed at reducing user effort and improving comfort.

Existing wheelchair technologies can be broadly categorized into manual, powered, and semi-automated systems. Manual wheelchairs rely entirely on human effort, either from the user or a caregiver, while powered wheelchairs incorporate electric motors and batteries to assist in movement. Semi-automated systems introduce limited features such as joystick-based navigation and basic electronic controls. These developments have significantly improved mobility compared to traditional methods; however, they still fall short in providing complete independence and intelligent assistance.

In real-world scenarios, wheelchair users often encounter complex environments such as crowded areas, uneven terrains, and narrow pathways. Navigating through such conditions requires continuous attention, physical effort, and precise control. Existing systems do not adequately support users in handling these challenges, especially those with severe physical impairments or limited motor control. As a result, users may experience difficulty in safe navigation and may remain dependent on external assistance.

Moreover, most conventional wheelchair systems lack integration with modern technologies such as sensors, wireless communication, and real-time monitoring systems. The absence of features like obstacle detection, automated navigation, and emergency alert mechanisms limits the safety and usability of these devices. In critical situations, such as sudden obstacles or medical emergencies, traditional systems fail to provide timely assistance or preventive measures.

Another important aspect is the lack of user-friendly interfaces in existing systems. Control mechanisms such as joysticks or mechanical levers may not be suitable for all users, particularly those with severe disabilities affecting hand movement or coordination. This creates a barrier to accessibility and reduces the effectiveness of the wheelchair as an assistive device.

Therefore, while existing wheelchair methods have contributed significantly to improving mobility, they still have several shortcomings in terms of intelligence, safety, adaptability, and ease of use. These limitations emphasize the need for advanced smart wheelchair systems that

incorporate modern technologies such as voice recognition, gesture control, sensor-based navigation, and IoT connectivity to provide a more efficient, safe, and user-centric solution.

3.2 Traditional Wheelchair Systems

Traditional wheelchairs are the most commonly used mobility devices and can be broadly classified into two types:

1. Manual Wheelchairs

Manual wheelchairs are operated by the user or a caregiver using physical force. The user moves the wheelchair by pushing the wheels using their hands.

Features:

- Simple mechanical design
- Lightweight and portable
- Low cost and easy to maintain
- Does not require electricity

2. Electric (Powered) Wheelchairs

Electric wheelchairs are powered by motors and controlled using a joystick or control panel.

Features:

- Battery-operated system
- Controlled using joystick
- Requires less physical effort
- Suitable for users with limited strength

Although electric wheelchairs provide better ease of use compared to manual ones, they still rely on physical interaction through joystick control, which may not be suitable for all users.

3.3 Limitations of Existing Methods

Despite their usefulness, traditional and existing wheelchair systems have several limitations:

1. Lack of Accessibility

Manual wheelchairs require significant physical effort, making them unsuitable for individuals with severe disabilities. Electric wheelchairs, although easier to use, still require hand movement for joystick control.

2. No Intelligent Control

Most existing systems do not support advanced control methods such as voice commands or gesture recognition. This limits usability for users with restricted motor skills.

3. Safety Issues

Traditional wheelchairs lack obstacle detection and collision avoidance systems. This increases the risk of accidents, especially in crowded or unfamiliar environments.

4. Dependency on Caregivers

Users often depend on others for movement and navigation, reducing independence and confidence.

5. Absence of Real-Time Monitoring

Existing wheelchairs do not provide IoT-based features such as remote monitoring, location tracking, or emergency alerts, which are crucial for user safety.

6. Limited Adaptability

Most wheelchairs are not customizable according to user needs and environmental conditions, leading to reduced efficiency and comfort.

7. High Cost of Advanced Models

Some modern powered wheelchairs with advanced features are expensive, making them inaccessible to many users.

CHAPTER 4: PROPOSED METHOD

4.1 Overview of the Proposed System

The proposed system is an advanced smart wheelchair designed to enhance mobility, safety, and independence for physically disabled individuals. The system integrates modern technologies such as voice recognition, gesture control, obstacle detection, and Internet of Things (IoT) to provide an intelligent and user-friendly mobility solution.

The core objective of the proposed method is to enable hands-free and efficient control of the wheelchair. Unlike traditional systems that rely on manual operation or joystick control, this system allows users to navigate using voice commands and hand gestures. A microphone-based voice recognition module captures user commands such as forward, backward, left, and right, while a gesture sensor or accelerometer detects hand movements and converts them into control signals. These inputs are processed by a microcontroller, which acts as the central unit of the system.

The microcontroller interprets the received inputs and generates appropriate signals to control the motors through a motor driver circuit. This ensures smooth and accurate movement of the wheelchair in different directions. The system is designed to prioritize user commands while maintaining safety and stability during operation.

To enhance safety, the proposed system incorporates an obstacle detection mechanism using ultrasonic or proximity sensors. These sensors continuously monitor the surroundings of the wheelchair and detect any obstacles in its path. When an obstacle is identified within a predefined distance, the system automatically stops or alters the movement to prevent collisions. This feature is particularly useful in crowded or unfamiliar environments.

In addition to mobility and safety, the system also integrates IoT functionality for real-time monitoring and communication. A wireless communication module, such as Wi-Fi or Bluetooth, is used to transmit data to a cloud platform or mobile application. This allows caregivers or family members to monitor the wheelchair’s status and receive alerts in case of emergencies or abnormal conditions.

The proposed method also emphasizes simplicity, cost-effectiveness, and ease of use. The system is designed using commonly available components, making it affordable and accessible for a wider range of users. The user interface is kept simple to ensure that individuals with minimal technical knowledge can operate the system without difficulty.

Overall, the proposed smart wheelchair system combines multiple control methods, safety

features, and connectivity options into a single integrated solution. It aims to overcome the limitations of existing systems by providing a reliable, intelligent, and user-centric mobility aid that significantly improves the quality of life for its users.

4.2 System Architecture

The system architecture of the proposed smart wheelchair is designed to integrate multiple functional units such as input, processing, output, safety, and communication modules. The architecture follows a modular approach, where each unit performs a specific task and interacts with other components to ensure smooth and efficient operation of the system.

The overall system is divided into three major sections:

- Input Unit
- Processing Unit
- Output Unit

In addition, safety and IoT communication modules are incorporated to enhance reliability and real-time monitoring.

All inputs from the sensors and control modules are sent to the processing unit, which consists of a microcontroller acting as the central brain of the system. The microcontroller processes the received signals based on predefined logic, interprets user commands, and prioritizes safety conditions. It ensures that obstacle detection data is continuously evaluated so that any potential collision is avoided before executing movement commands. Once the decision-making process is completed, the microcontroller sends appropriate control signals to the output unit.

The output unit includes a motor driver and DC motors that are responsible for the physical movement of the wheelchair. The motor driver serves as an interface between the microcontroller and the motors by amplifying the control signals to the required level. Based on these signals, the DC motors rotate in specific directions, enabling the wheelchair to move forward, backward, or turn as per user input. The safety mechanism is integrated within the system architecture, where the wheelchair automatically stops or alters its direction when an obstacle is detected within a predefined distance, thereby preventing accidents and ensuring user protection.

In addition to movement and safety, the system incorporates an IoT communication module to enhance functionality and connectivity. A wireless communication unit, such as Wi-Fi or Bluetooth, allows the system to transmit data to a cloud platform or mobile application. This enables real-time monitoring of the wheelchair’s status and provides alerts in case of

emergencies or abnormal conditions. Overall, the system architecture seamlessly integrates multiple components to create a smart, efficient, and user-friendly wheelchair system that improves mobility, safety, and independence for the user.

Fig:4.1 Flow chart of the System

4.3 Block Diagram of the System

The block diagram represents a smart wheelchair system centered around the ESP32 microcontroller, which acts as the main processing unit. It receives input from multiple sources such as the accelerometer for gesture control, the voice recognition module for voice commands, and ultrasonic sensors for obstacle detection. The ESP32 processes these inputs and accordingly sends control signals to the motor driver, which operates the DC motors to move the wheelchair in different directions. The system is powered by a battery, while a regulated power supply ensures proper functioning of components. Additionally, the ESP32 connects to the Blynk IoT cloud through Wi-Fi, allowing the wheelchair to be monitored and controlled via a mobile phone. A buzzer is also included to provide alerts for safety or warning conditions, making the system more reliable and user-friendly.

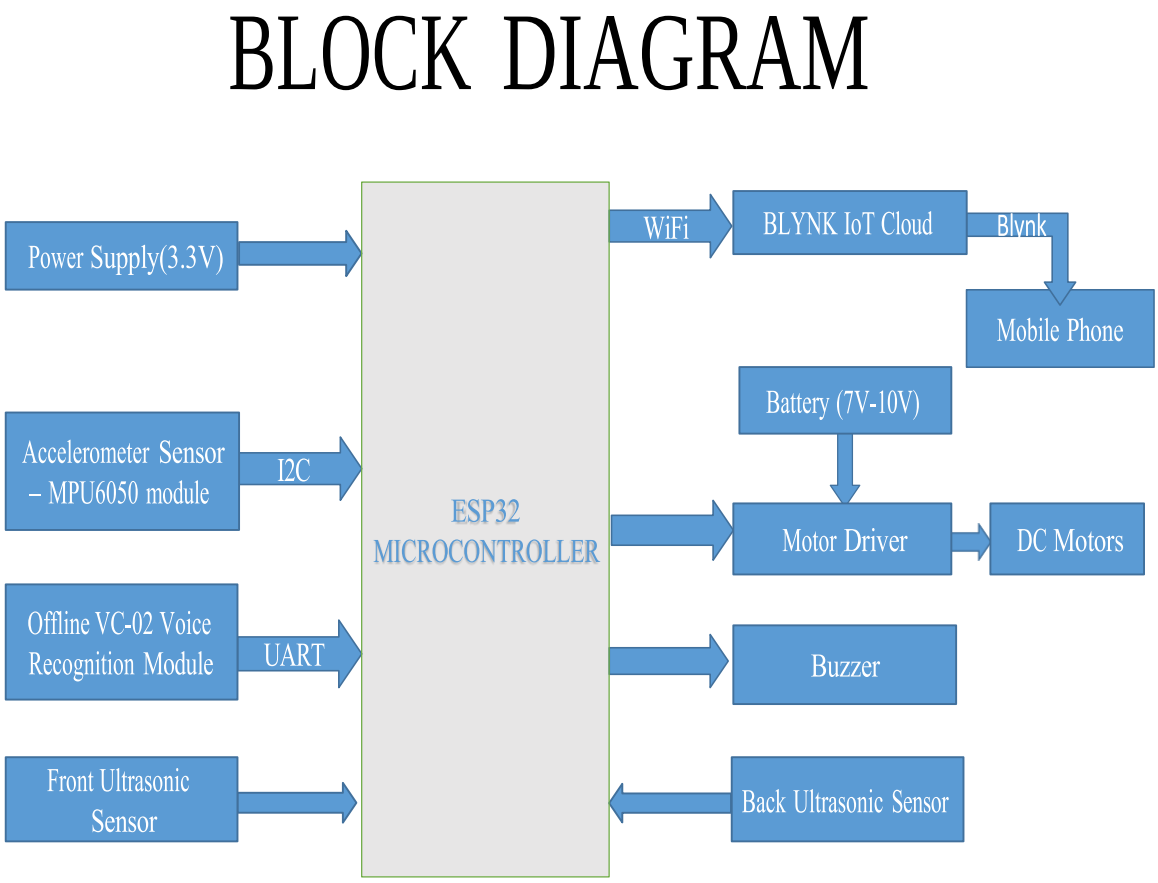


Fig:4.2 Block Diagram of the system

4.4 Hardware Components Description

The major hardware components used in the system include:

- **ESP32 Microcontroller**

  The ESP32 acts as the main processing unit of the smart wheelchair system, controlling all components and coordinating their operations. It is equipped with a dual-core processor and supports Free RTOS, which allows it to perform multiple tasks simultaneously such as reading sensor data, processing user commands, and controlling motor functions. The ESP32 receives inputs from the voice recognition module, accelerometer, and ultrasonic sensors, and processes them in real time to ensure smooth and accurate wheelchair movement while prioritizing safety.

  In addition, the ESP32 has built-in Wi-Fi and Bluetooth capabilities, enabling seamless IoT integration. It connects to cloud platforms like Blynk to provide real-time monitoring and control through a mobile phone. The microcontroller also supports multiple communication protocols such as UART and I2C, making it easy to interface with different modules. Its fast-processing speed, low power consumption, and versatility make it highly suitable for implementing an efficient and intelligent smart wheelchair system.

Fig:4.3 ESP 32 Module

- **MPU6050 Sensor**

  The MPU6050 is a motion tracking sensor that combines both an accelerometer and a gyroscope in a single module, enabling it to measure linear acceleration and angular velocity along multiple axes. In the smart wheelchair system, it is used to detect hand movements or tilts, which are interpreted as gesture-based commands. These gestures are then converted into directional instructions such as forward, backward, left, or right, allowing intuitive and hands-free control of the wheelchair.

  The sensor communicates with the microcontroller (ESP32) using the I2C protocol, providing continuous real-time data for accurate motion detection. By analyzing this data, the system can distinguish between different gestures and ensure smooth and responsive control. The MPU6050 is compact, cost-effective, and energy-efficient, making it suitable for wearable or handheld applications in assistive systems like smart wheelchairs.

Fig:4.4 MPU6050

- **VC-02 Voice Recognition Module**

This voice recognition module is capable of identifying predefined voice commands without the need for an internet connection, making it reliable and efficient for offline operation. It processes spoken inputs locally and converts them into digital signals that can be understood by the microcontroller. This ensures quick response time and reduces dependency on network availability.

In the smart wheelchair system, the module enables users to control movement using simple voice instructions such as forward, backward, left, and right. This feature is especially beneficial for individuals with limited physical mobility, as it provides a hands-free and user-friendly method of controlling the wheelchair, improving accessibility and independence.

Fig:4.5 VC-02 Voice Recognition Module

- **Ultrasonic Sensors (Front & Back)**

Ultrasonic sensors are placed at the front and back of the wheelchair to measure the distance between the system and nearby obstacles. They work by emitting ultrasonic waves and calculating the time taken for the echo to return, which helps determine the distance of objects in the path. This continuous sensing allows the system to be aware of its surroundings in real time.

In the smart wheelchair, these sensors play a crucial role in ensuring safety by enabling collision avoidance. When an obstacle is detected within a certain threshold distance, the system automatically stops or restricts movement to prevent accidents. This feature enhances user confidence and allows safer navigation, especially in crowded or unfamiliar environments.

Fig:4.6 Ultrasonic Sensor

- **Motor Driver (L293D)**

The L293D motor driver acts as an interface between the ESP32 microcontroller and the DC motors of the wheelchair. Since the ESP32 cannot supply sufficient current to drive motors directly, the motor driver amplifies the control signals and provides the required power to operate the motors efficiently. It receives input signals from the controller and converts them into appropriate outputs for motor operation.

In the smart wheelchair system, the L293D controls both the direction and speed of the motors based on the commands processed by the ESP32. By switching the polarity of the motor inputs, it enables forward, backward, left, and right movements. This ensures smooth and precise control of the wheelchair, making navigation easier and more reliable for the user.

Fig:4.7 Motor Driver

- **DC Motors**

DC motors are the primary actuators responsible for the movement of the wheelchair. They convert electrical energy supplied by the battery into mechanical motion, enabling the wheels to rotate. By controlling the rotation of these motors, the wheelchair can move forward, backward, and perform turning operations based on user commands.

In the system, the motors are driven through the motor driver, which regulates their direction and speed according to signals from the ESP32. By varying the speed of individual motors, the wheelchair can achieve smooth turning and precise navigation. Their reliable performance and quick response make DC motors suitable for controlled and efficient mobility in the smart wheelchair.

Fig:4.8 DC Motors

- **Buzzer**

The buzzer provides an audio alert when an obstacle is detected or during emergency situations, ensuring user awareness.

Fig:4.9 Buzz

- **Power Supply (Battery)**

The lithium battery supplies the required electrical power to operate the entire smart wheelchair system, ensuring portability and continuous functioning. Lithium batteries are preferred due to their high energy density, lightweight nature, and longer lifespan compared to conventional batteries. They provide a stable and efficient power source for components such as the ESP32, sensors, motor driver, and DC motors.

In this system, the lithium battery ensures uninterrupted operation and supports longer usage time with faster charging capability. Its compact size makes it suitable for integration into the wheelchair without adding significant weight. Additionally, lithium batteries offer better efficiency and reliability, making them ideal for modern IoT-based assistive devices.

Fig:4.10 Power supply

- **Blynk IoT Platform**

Blynk is an IoT platform used to remotely monitor and control the smart wheelchair through a mobile application. It connects with the ESP32 via Wi-Fi and provides a user-friendly dashboard where important system parameters can be viewed in real time. This allows seamless communication between the wheelchair and the user or caregiver.

In this system, Blynk displays information such as movement status, sensor readings, and obstacle distances, helping users track the wheelchair's operation continuously. It can also be configured to send alerts during abnormal conditions, enhancing safety and reliability. This remote monitoring feature adds convenience and ensures better supervision of the wheelchair's performance.

Fig:4.11 Blynk IoT Platform

4.5 Software Requirements

The software components required include:

- Arduino IDE for programming
- Embedded C/C++ for coding
- Libraries for sensor and module integration
- Serial communication tools for debugging
- UniOneUpdate
- Ai-Thinker

The software components required for the smart wheelchair system include a combination of development tools, programming languages, and supporting libraries that enable efficient system design and implementation. The Arduino IDE serves as the primary platform for writing, compiling, and uploading code to the ESP32 microcontroller. Embedded C/C++ is used as the programming language, providing low-level control over hardware components and ensuring fast and reliable execution of tasks. To simplify development, various libraries are integrated for handling sensors such as the MPU6050, ultrasonic sensors, and communication modules, reducing complexity and improving code efficiency.

In addition to programming, serial communication tools play a crucial role in debugging and real-time monitoring of data, allowing developers to track sensor outputs and system behavior during operation. UniOneUpdate software is used to train and upload predefined commands to the voice recognition module, enabling accurate voice-based control. Ai- Thinker tools are utilized for configuring the ESP32, managing firmware, and ensuring proper Wi-Fi connectivity for IoT applications. These tools help in establishing a stable connection with platforms like Blynk for remote monitoring. Overall, the combination of these software components ensures smooth system development, reliable performance, easy troubleshooting, and efficient integration of all hardware modules in the smart wheelchair system.

CHAPTER 5: SYSTEM IMPLEMENTATION AND WORKING

5.1 SYSTEM WORKING

The proposed smart wheelchair system operates using the ESP32 microcontroller as the central controller, which manages all sensors, modules, and communication processes. The system is designed using a Real-Time Operating System (RTOS), allowing multiple tasks to run simultaneously such as voice processing, gesture control, obstacle detection, and IoT communication. This ensures fast response, smooth operation, and efficient handling of different functionalities.

1. Voice Control

The VC02 offline voice recognition module receives voice commands such as:

- Forward
- Backward
- Left
- Right
- Stop
- Emergency
- Turn ON Light
- Turn OFF Light

The voice control mechanism is implemented using the VC02 offline voice recognition module, which detects predefined commands such as forward, backward, left, right, stop, emergency, and light control commands. These commands are transmitted to the ESP32 through serial communication, where they are processed and converted into motor control actions and auxiliary functions. Voice control is given the highest priority in the system to ensure immediate response to user commands. When an emergency command is detected, the system immediately stops all motor operations, updates the wheelchair status to “stopped,” and sends an alert notification to the Blynk platform. This feature is crucial for handling critical situations quickly and safely.

1. Gesture Control

- The MPU6050 sensor detects hand tilt and motion.
- Based on tilt direction:
  - Tilt Forward → Wheelchair moves forward
  - Tilt Back → Wheelchair moves backward
  - Tilt Left → Wheelchair turns left
  - Tilt Right → Wheelchair turns right
  - No tilt → Wheelchair stops

Gesture control is achieved using the MPU6050 sensor, which detects hand tilt and motion. Based on the direction of tilt, the system generates movement commands such as forward, backward, left, or right. If no tilt is detected, the wheelchair remains stationary. This control method works only when voice commands are not active, ensuring proper coordination between input methods and avoiding conflicts. Gesture control provides an alternative hands-free operation for users who may prefer motion-based interaction.

2. Obstacle Detection

- Two ultrasonic sensors are used:
- Front Ultrasonic Sensor
- Detects obstacles in front of wheelchair.
- If distance < 20 cm:
  - Forward motion stops
  - Left motion stops
  - Right motion stops
  - Motors stop immediately
  - Wheelchair status shows STOPPED in Blynk
- This prevents collision with objects in front.
- Back Ultrasonic Sensor
- Detects obstacles behind wheelchair.
  - If distance < 20 cm:
  - Backward motion stops
  - Motors stop immediately
  - Wheelchair status shows STOPPED
- This prevents collision while moving backward.

The system uses two ultrasonic sensors placed at the front and back of the wheelchair to continuously monitor the surroundings. These sensors measure the distance between the wheelchair and nearby obstacles.

If an obstacle is detected within a threshold distance (typically 20 cm), the system automatically restricts movement in that direction. For example, a front obstacle stops forward and turning movements, while a rear obstacle restricts backward movement. This ensures collision avoidance and safe navigation in different environments.

3. Buzzer Alert System

- The buzzer is used for safety warning.
- Buzzer Logic
- If:
- Front obstacle < 20 cm
  OR
  Back obstacle < 20 cm
- Then:
- Buzzer ON
- Even if wheelchair is not moving, buzzer will still sound.
- This continuously alerts the user that an obstacle is nearby.
- If no obstacle:
- Buzzer OFF
- This ensures safety at all times.

4. Blynk Dashboard Monitoring

- The system is connected to Blynk IoT platform through WiFi.
- The Blynk dashboard shows real-time data.
- V0 – Wheelchair Status
- Displays current movement of wheelchair:
  - Forward
  - Backward
  - Left
  - Right
  - Stopped

The system is connected to the Blynk IoT platform through Wi-Fi, enabling real-time monitoring via a mobile application. The dashboard displays important system parameters such as the current movement status of the wheelchair and the distance readings from both front and rear ultrasonic sensors.

This allows users or caregivers to remotely monitor the wheelchair's operation and receive updates or alerts during abnormal conditions, improving overall system reliability and supervision.

V1 – Front Obstacle Distance
Displays distance of front ultrasonic sensor in centimeters.
Example:
52 cm
10 cm
18 cm
This helps monitor front obstacles.
V2 – Back Obstacle Distance
Displays distance of back ultrasonic sensor.
Example:
9 cm
40 cm
100 cm
This helps monitor rear obstacles.

5.2 Safety Logic

The safety logic of the proposed smart wheelchair system is designed to ensure secure and reliable operation under all conditions. It continuously monitors data from the ultrasonic sensors and applies predefined rules to prevent accidents and unsafe movements. The system prioritizes user safety over movement commands, meaning that even if a valid command is given, it will not be executed if it leads to a potential collision.
When an obstacle is detected in front of the wheelchair within a threshold distance (approximately 20 cm), the system immediately restricts forward movement as well as left and right turning motions, since these directions may lead the wheelchair closer to the obstacle. Similarly, when an obstacle is detected at the rear side, backward movement is disabled to prevent collisions while reversing. In both cases, the motors are either stopped completely or controlled in a way that avoids unsafe directions.

In addition to movement restriction, the system activates the buzzer whenever an obstacle is detected, providing an immediate audio warning to the user. This alert works continuously until the obstacle is cleared, ensuring constant awareness of surrounding hazards. At the same time, the system updates the wheelchair status on the Blynk dashboard, allowing remote users or caregivers to monitor safety conditions in real time.
The safety logic also works in coordination with the voice control system, especially during emergency situations. If an emergency command is received, all operations are instantly halted regardless of sensor inputs, and an alert is sent through the IoT platform. Overall, this layered safety mechanism combining sensor data, alerts, and control restrictions ensures maximum protection and reliability during wheelchair operation.

The system ensures safety using multiple conditions.
Condition 1:
If front obstacle < 20 cm
Forward, Left, Right motion stopped
Condition 2:
If back obstacle < 20 cm
Backward motion stopped
Condition 3:
If obstacle detected
Buzzer ON
Condition 4:
Wheelchair status updated in Blynk

5.3 RTOS Task Structure

The ESP32 runs multiple tasks simultaneously.
Task 1 – Voice Control (Highest Priority)
Handles voice commands and emergency stop.
Runs continuously.
Task 2 – Obstacle Detection
Reads ultrasonic sensors and controls buzzer and motor safety.
Updates Blynk dashboard.
Task 3 – Blynk Communication
Maintains WiFi and Blynk connection.
Sends data to dashboard.
Task 4 – Gesture Control
Reads MPU6050 and controls movement.
Runs only when voice control is inactive.

The system utilizes the Real-Time Operating System (RTOS) available in the ESP32 to manage multiple operations efficiently and simultaneously. Instead of executing tasks sequentially, RTOS allows the system to run multiple tasks in parallel with proper scheduling and prioritization. This is essential for real-time applications like a smart wheelchair, where quick

response and coordination between different modules are critical.

The voice control task is assigned the highest priority because it directly handles user commands and emergency instructions. This ensures that voice inputs are processed immediately without delay. The obstacle detection task runs continuously in the background, reading sensor data and enforcing safety logic such as stopping the motors or activating the buzzer when required. This task is crucial for real-time safety monitoring.

The Blynk communication task is responsible for maintaining Wi-Fi connectivity and updating data to the cloud platform. It ensures that system parameters such as wheelchair status and obstacle distances are transmitted regularly without interrupting other operations. The gesture control task operates with lower priority and is activated only when voice control is not in use. It reads data from the MPU6050 sensor and converts gestures into movement commands.

RTOS scheduling ensures that each task gets sufficient processing time based on its priority, preventing delays and conflicts. This structured task management improves system responsiveness, stability, and efficiency. It also allows the system to handle multiple inputs and operations simultaneously, making the smart wheelchair more reliable and effective in real-time environments.

CHAPTER 6: EXPERIMENTAL RESULTS

The proposed smart wheelchair system was tested for voice control, gesture control, obstacle detection, and IoT communication. The voice recognition module accurately detected predefined commands in a noise-free environment, and the emergency command successfully stopped the wheelchair immediately. Gesture control using the MPU6050 responded correctly to hand movements, though minor inaccuracies were observed due to sensor sensitivity.

Fig:6.1 VOICE COMMAND: EMERGENCY -> Emergency notification sent to mobile phone

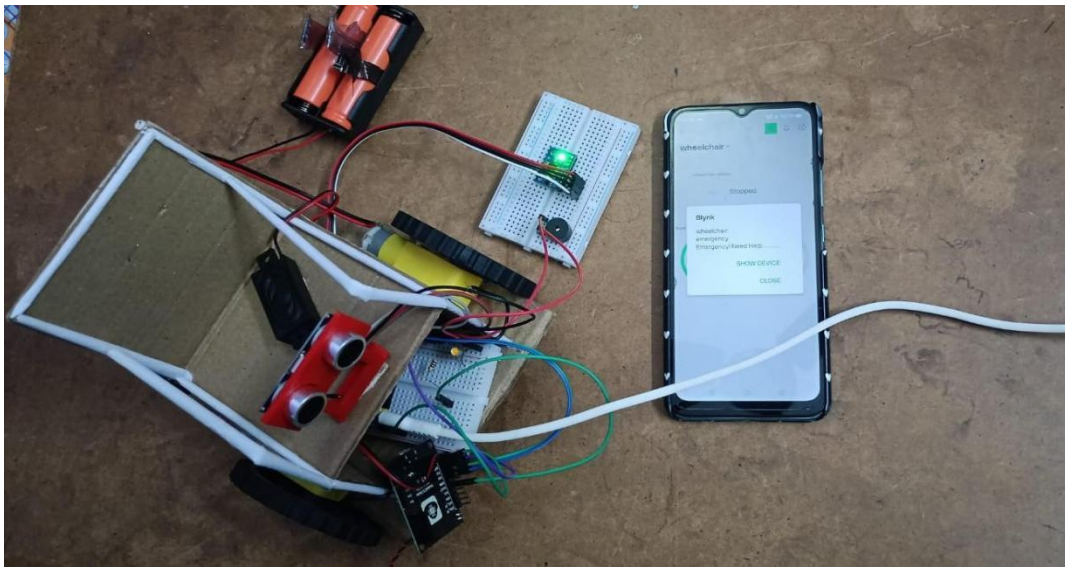


Fig:6.2 VOICE COMMAND: Turn on the Light

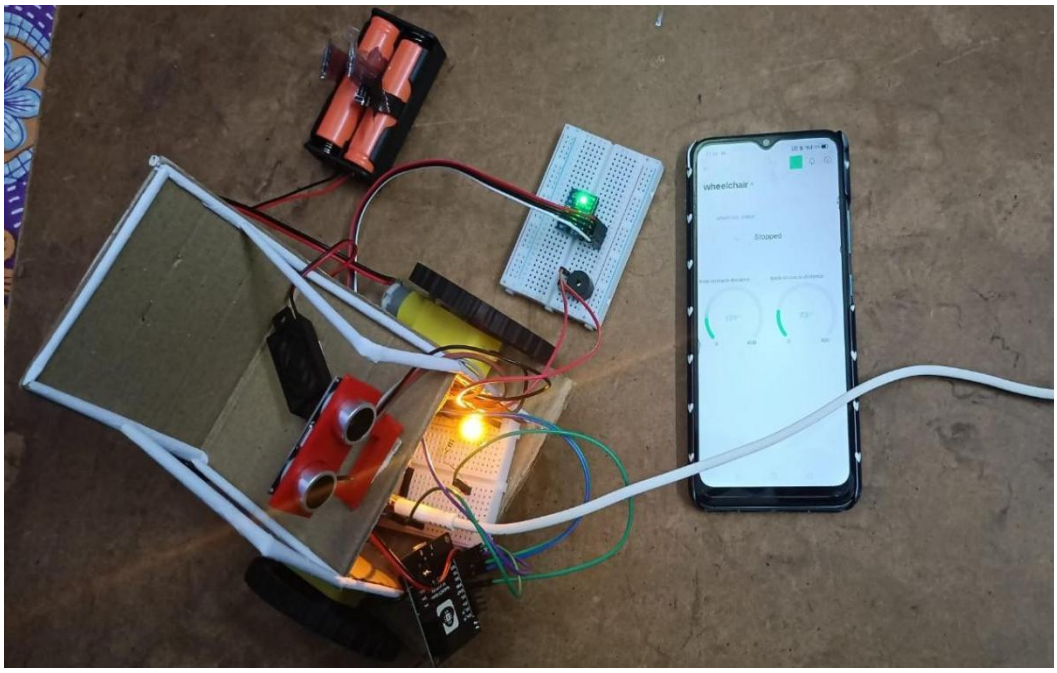


Fig:6.3 VOICE COMMAND: Turn off the Light

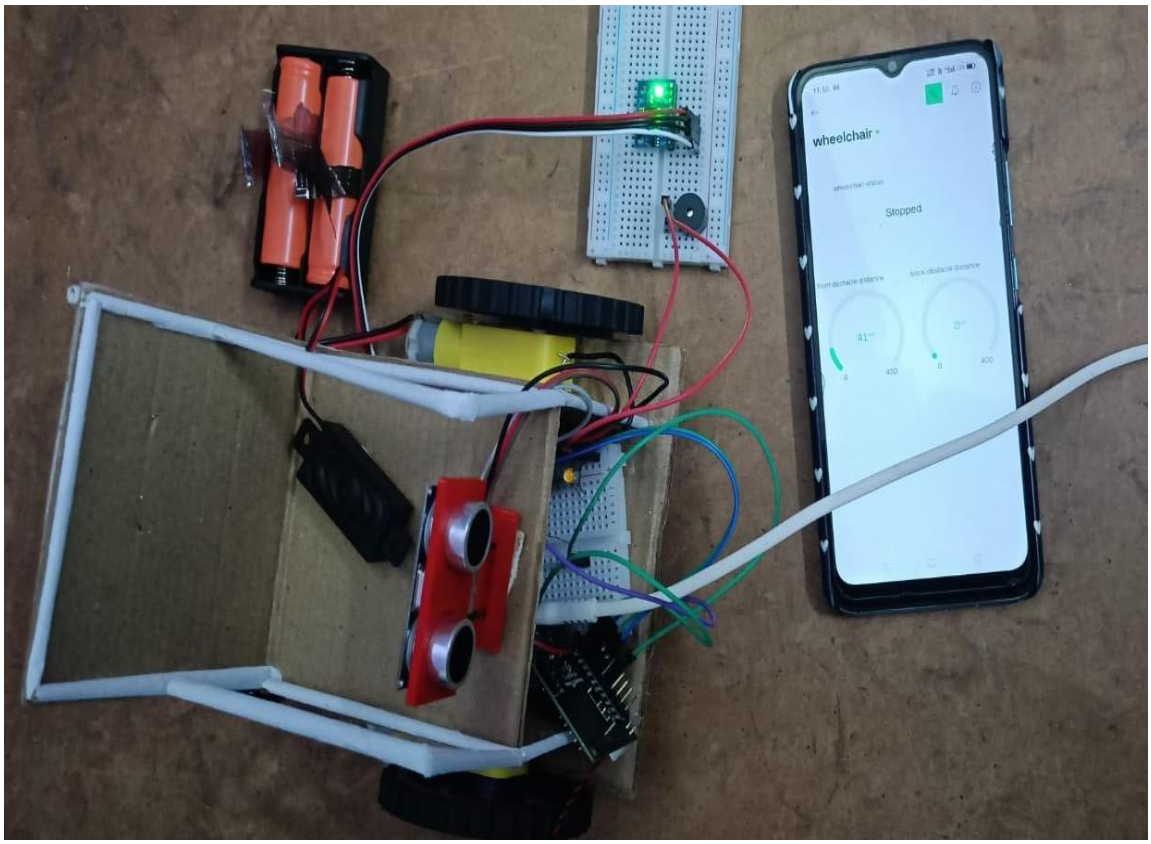


Fig:6.4 VOICE COMMAND: STOP

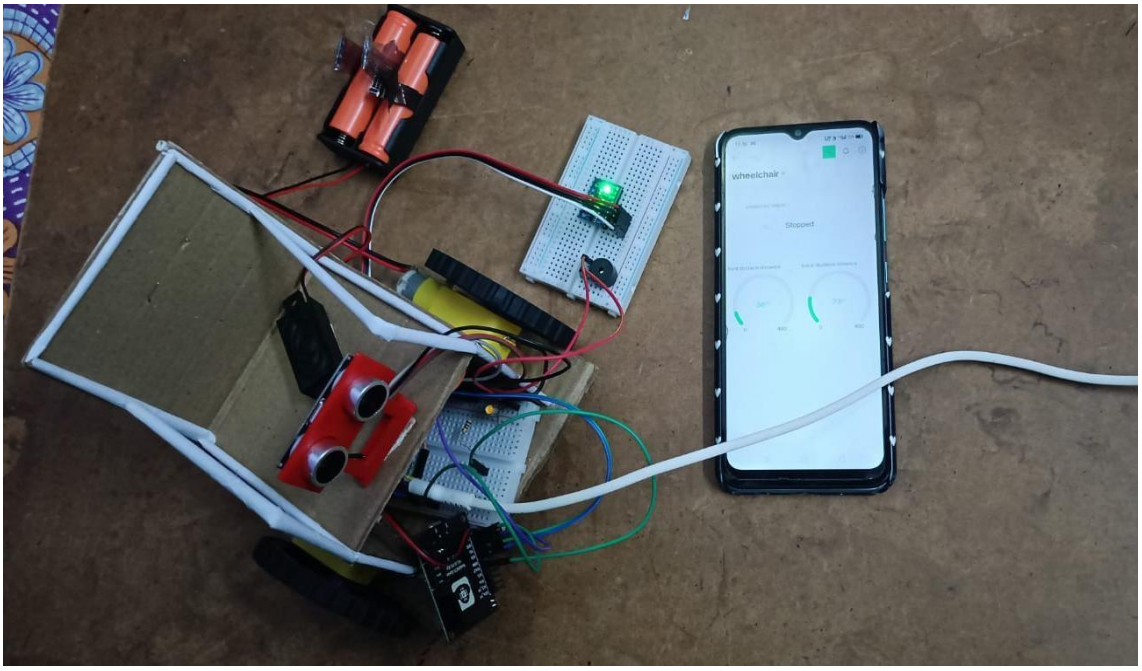


Fig:6.5 VOICE COMMAND: FORWARD

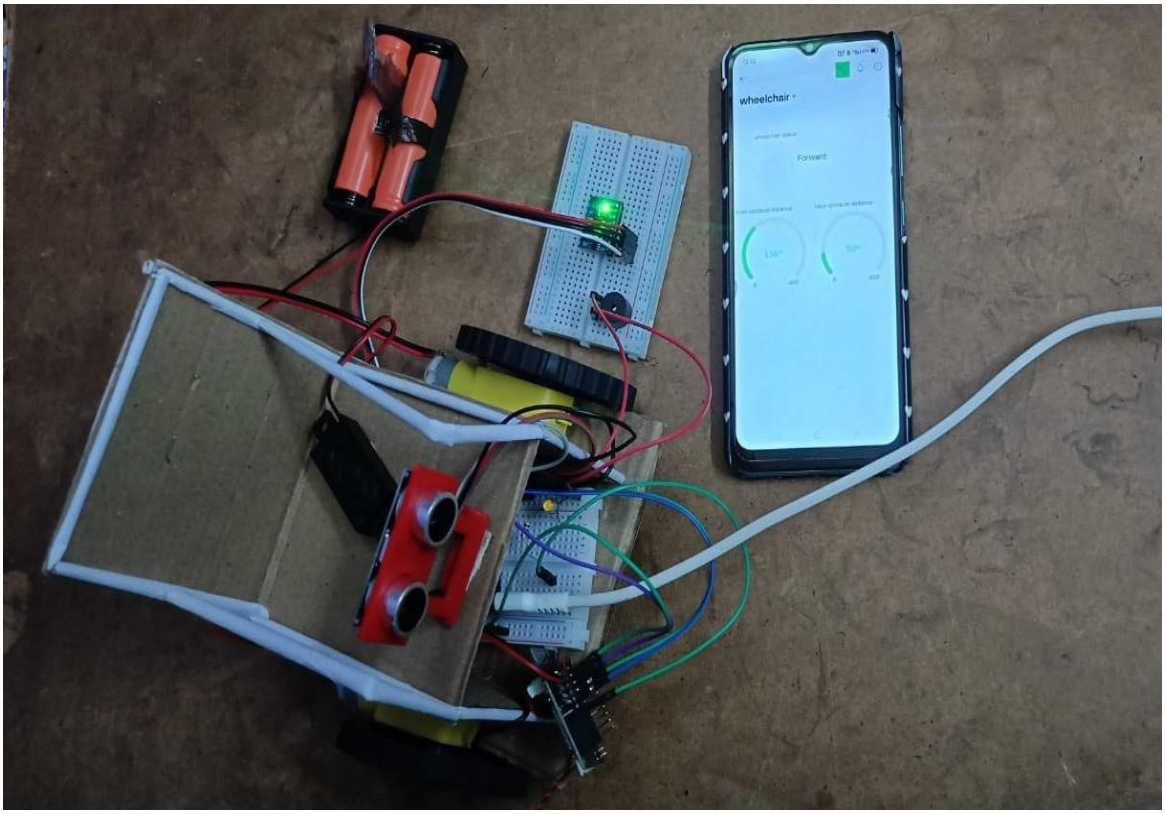


Fig:6.6 VOICE COMMAND: BACKWARD

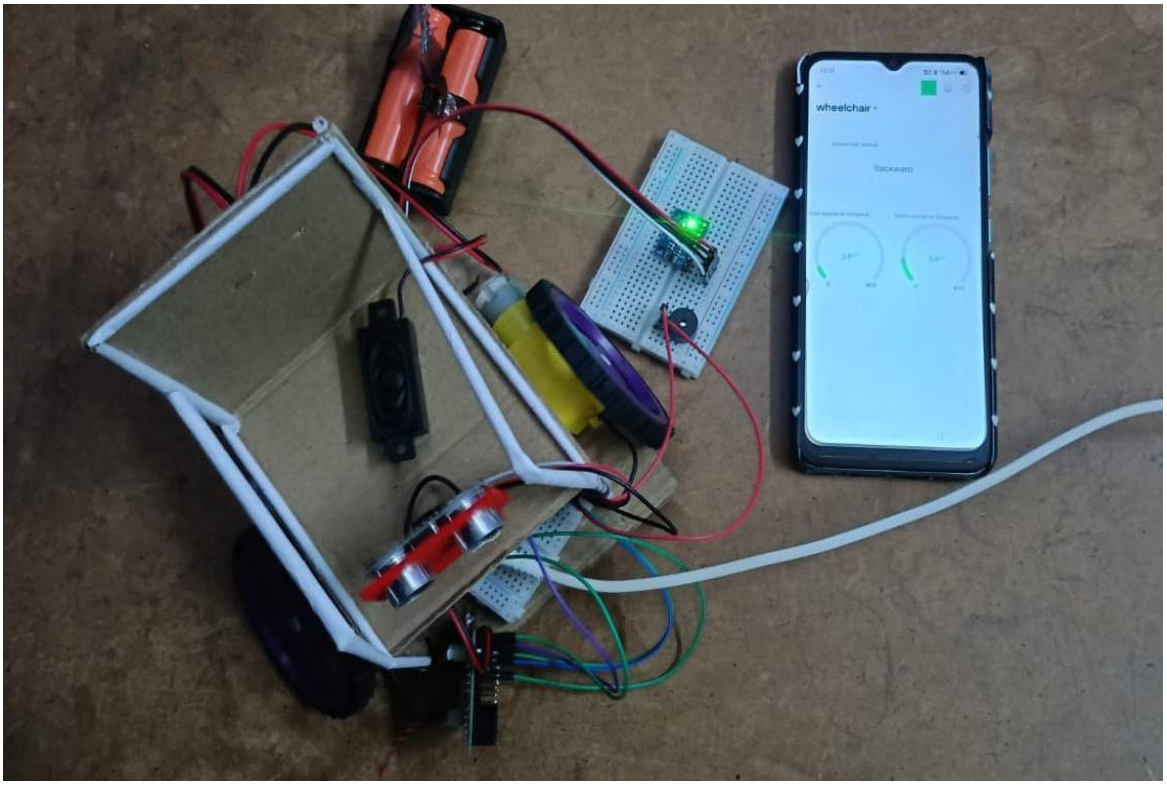


Fig:6.7 VOICE COMMAND: LEFT

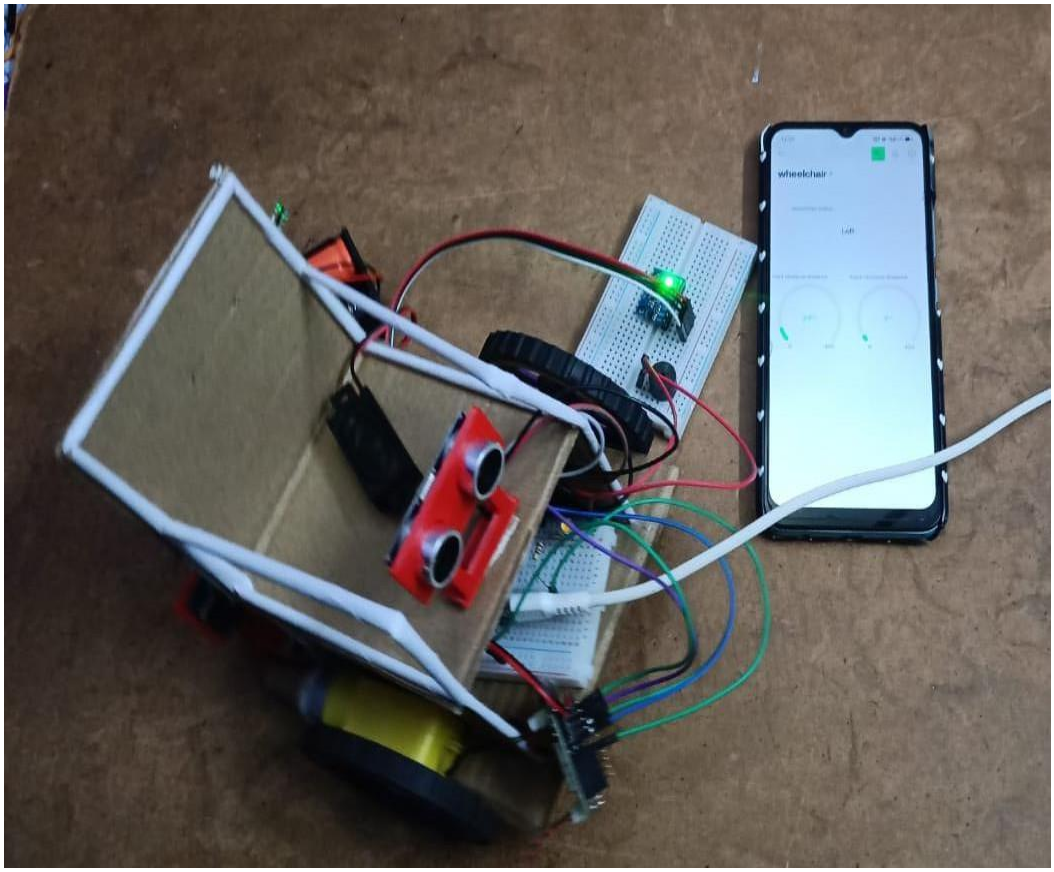


Fig:6.8 VOICE COMMAND: RIGHT

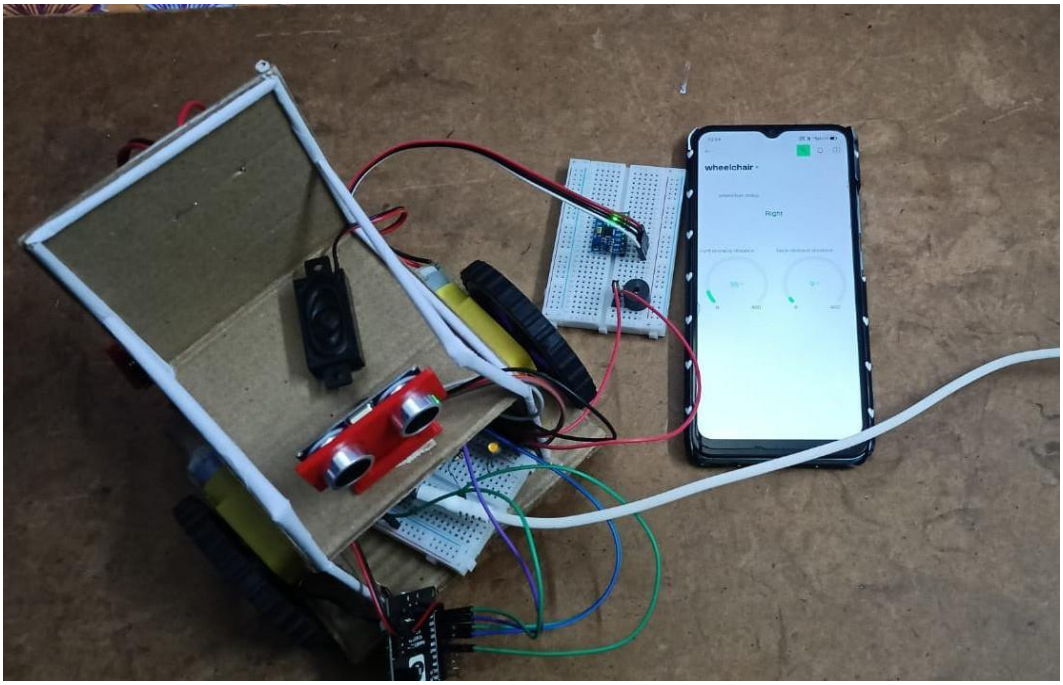


Fig:6.9 GESTURE: STOP

Fig:6.10 GESTURE: FORWARD

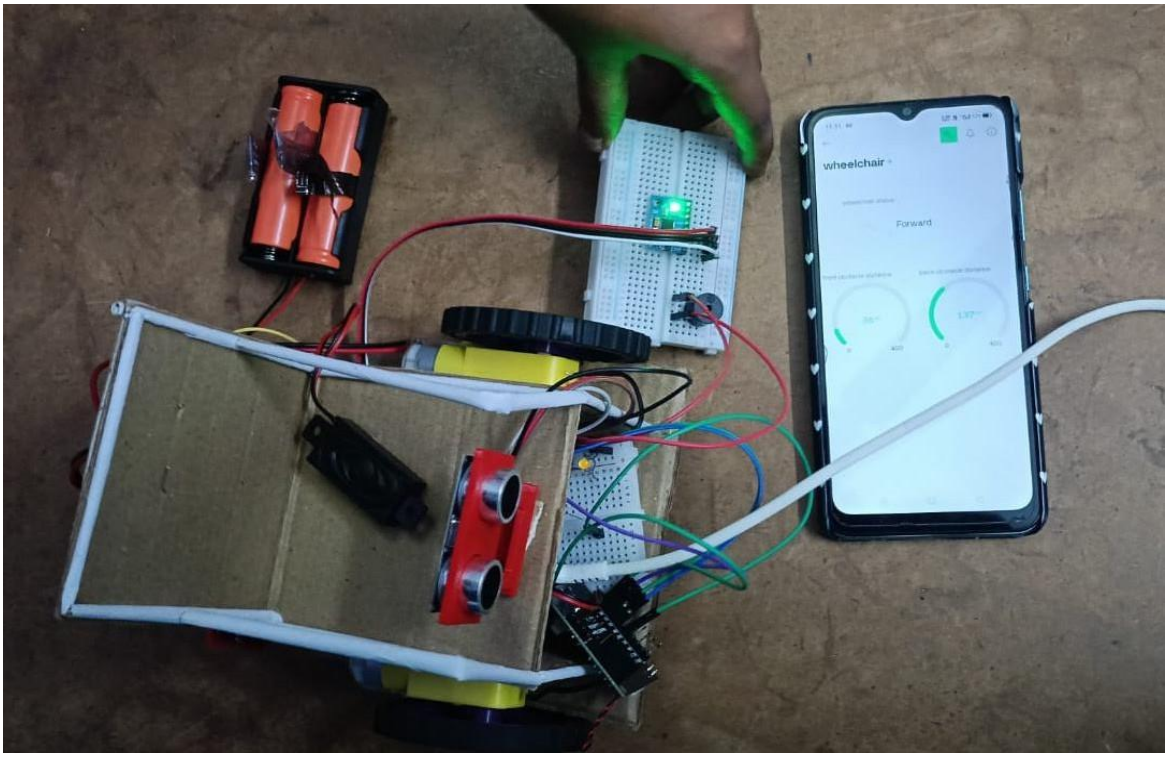


Fig:6.11 GESTURE: BACKWARD

Fig:6.12 GESTURE: RIGHT

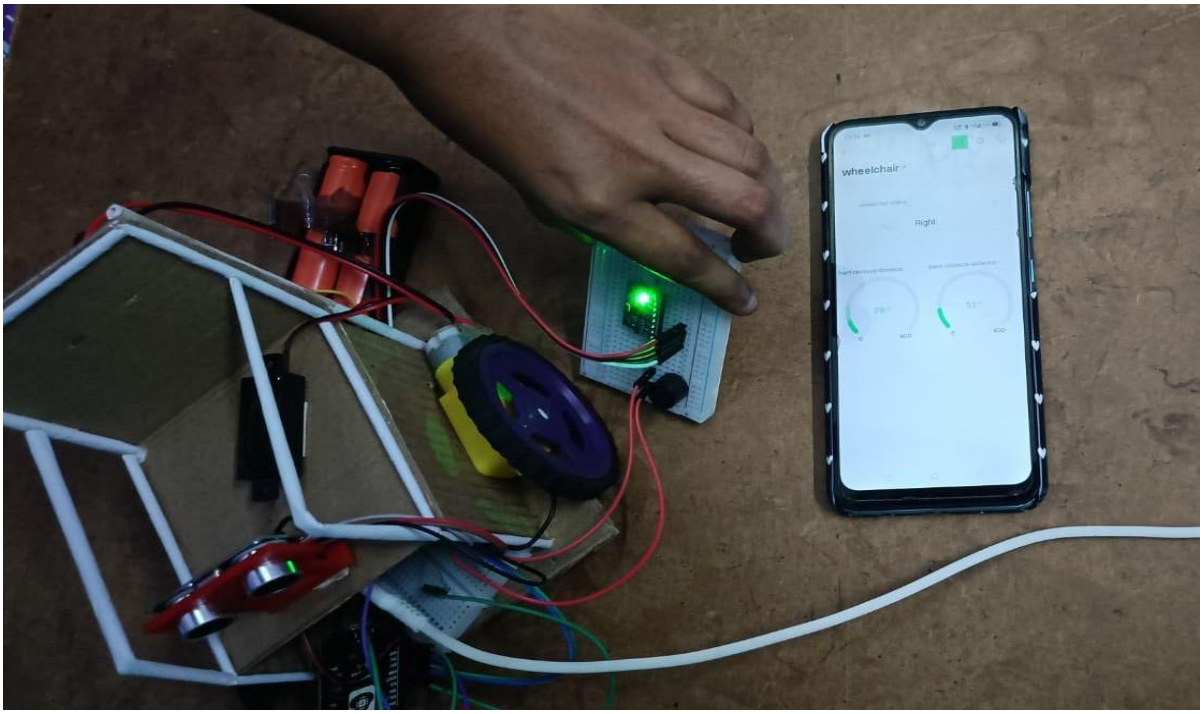


Fig:6.13 GESTURE: LEFT

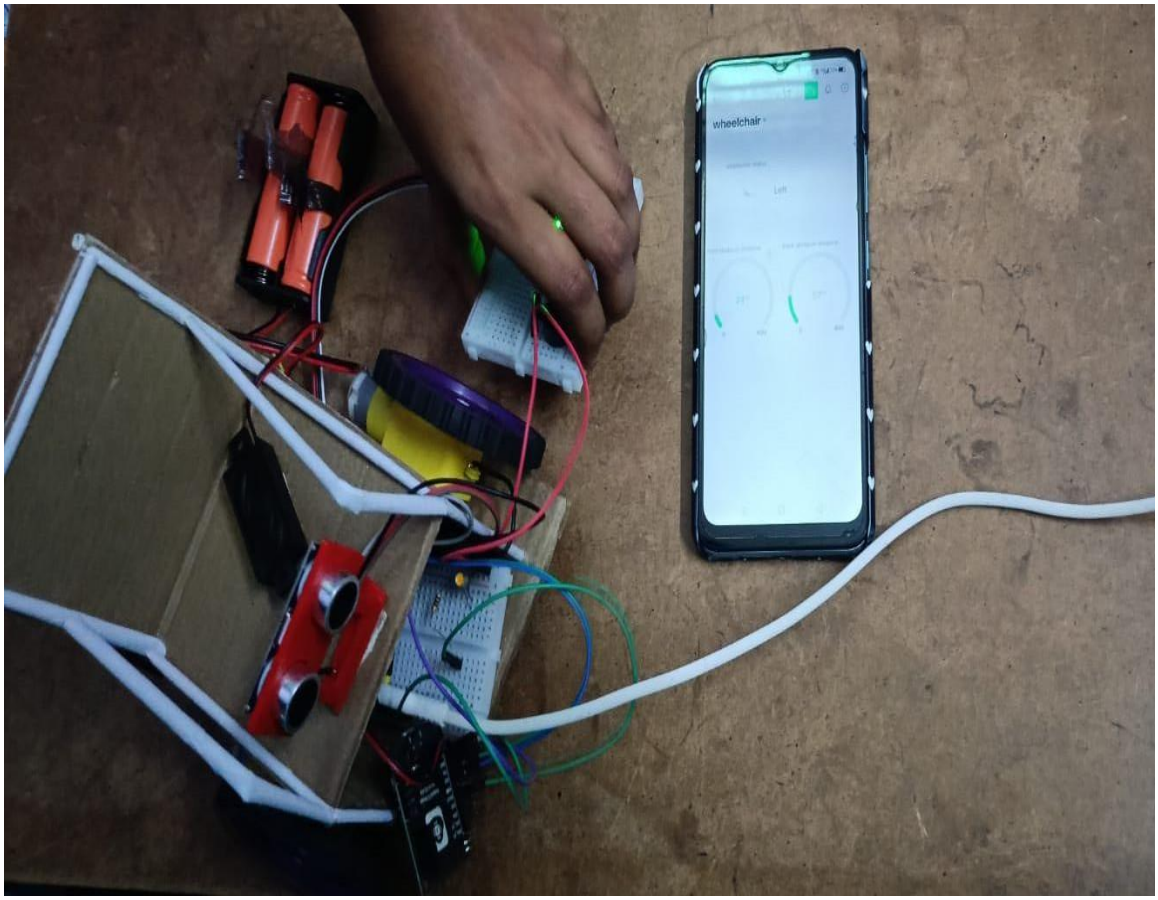


Fig:6.14 Front obstacle-> distance is shown in blynk buzzer on (<threshold=20 cm)

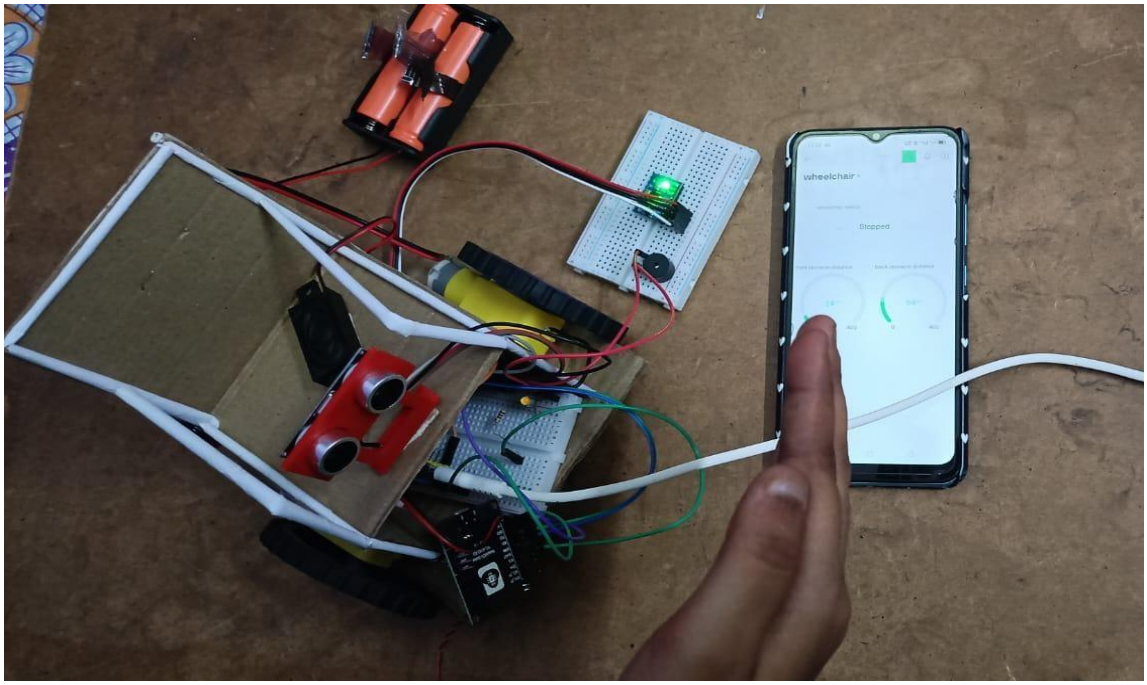


Fig:6.15 Back Obstacle -> distance is shown in blynk buzzer on(<threshold=20cm)

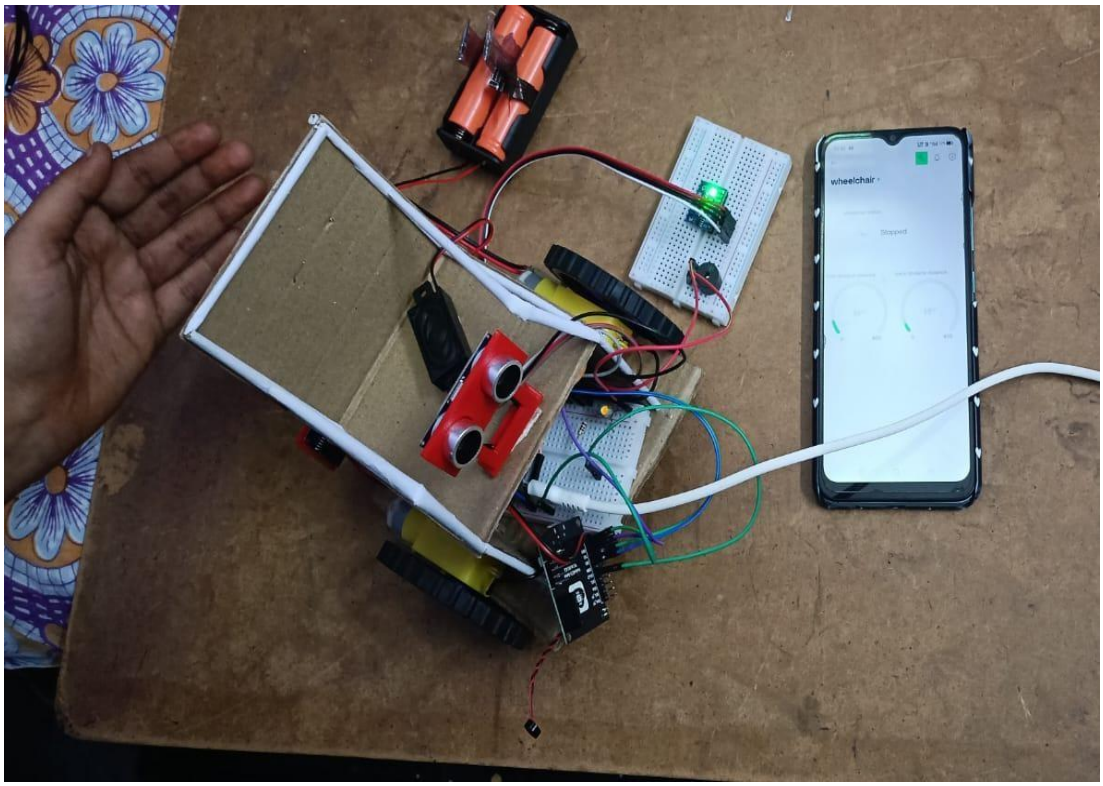


Fig:6.16 BackObstacle -> distance is shown in blynk (>threshold=20cm)

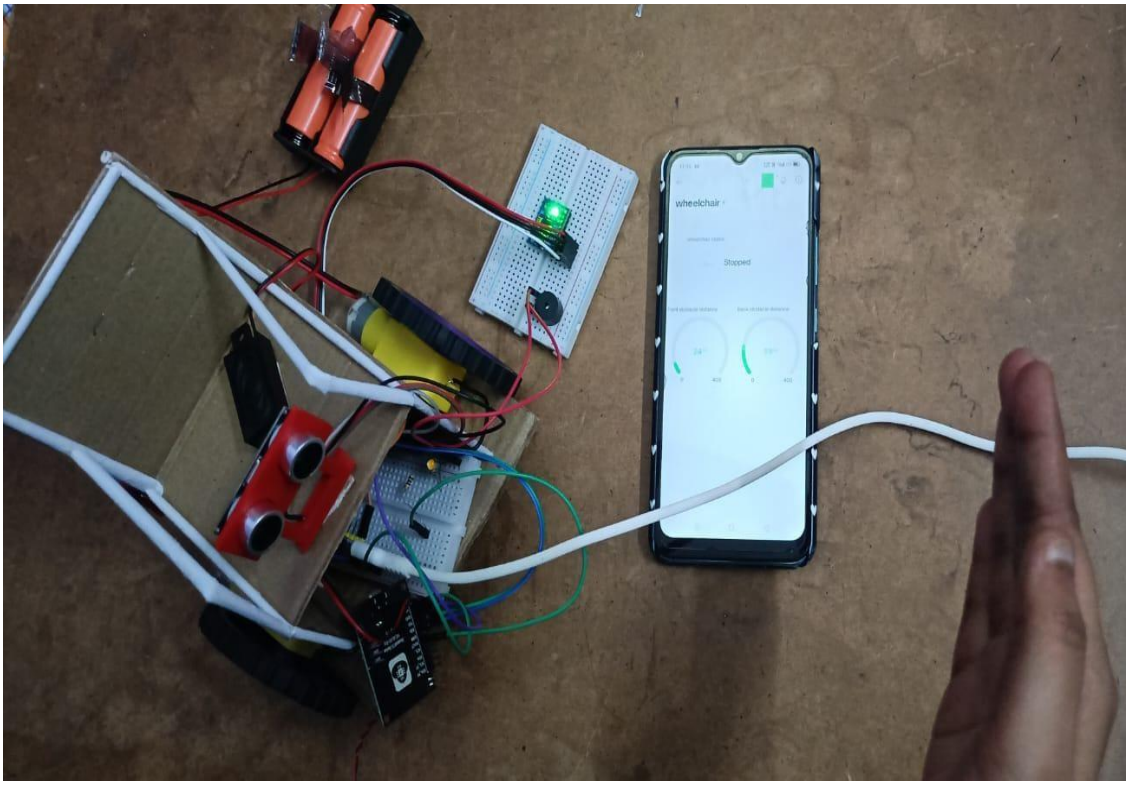
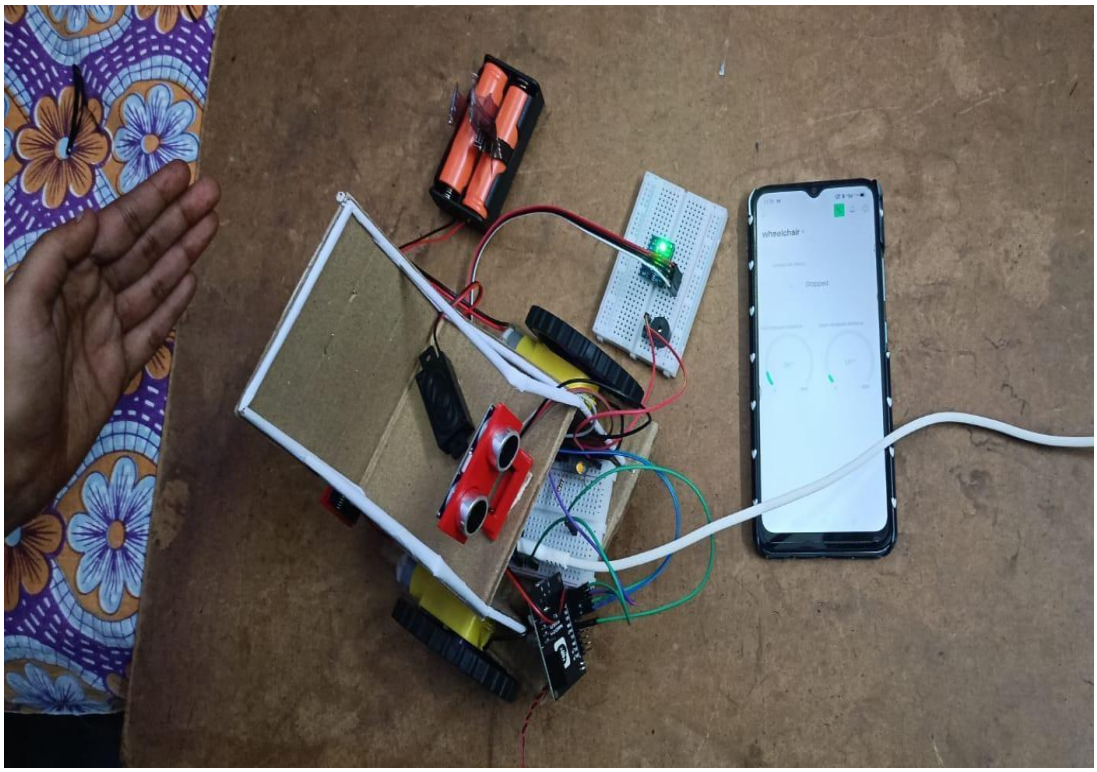


Fig:6.17 Front Obstacle -> distance is shown in blynk (>threshold=20cm)



The ultrasonic sensors effectively detected obstacles within 20 cm and prevented collisions by stopping the motors, while the buzzer provided timely alerts. The Blynk IoT platform displayed real-time data such as movement status and obstacle distances with minimal delay. The RTOS-based system ensured smooth multitasking and fast response. Overall, the system performed reliably and achieved the intended objectives with good accuracy and safety.

CHAPTER 7: DISCUSSION OF RESULTS

The implementation of the proposed smart wheelchair system demonstrates significant improvements in mobility assistance, user interaction, and safety when compared to conventional wheelchair systems. The integration of multiple control methods, including voice and gesture recognition, allows flexible and user-friendly operation. During testing, the voice control module successfully recognized predefined commands with good accuracy in normal indoor environments, enabling smooth and responsive wheelchair movement. The priority-based control ensured that voice commands, especially emergency instructions, were executed immediately without delay, enhancing system reliability in critical situations.

The gesture control system using the MPU6050 sensor also showed effective performance in detecting hand tilts and translating them into directional movement. It provided an alternative control method for users who may not be able to use voice commands. However, slight variations in hand movement and sensor sensitivity sometimes resulted in minor inaccuracies, indicating the need for calibration and filtering techniques for improved precision. Despite this, the combined use of voice and gesture control increased overall system flexibility and usability.

The obstacle detection mechanism using ultrasonic sensors played a crucial role in ensuring user safety. The sensors were able to detect obstacles within the defined threshold distance and successfully prevented collisions by stopping or restricting movement. The system responded quickly to obstacles in both forward and backward directions, demonstrating effective real-time performance. The buzzer alert system further enhanced safety by continuously notifying the user about nearby obstacles, even when the wheelchair was stationary.

The IoT integration using the Blynk platform proved to be highly effective for remote monitoring. The system was able to transmit real-time data such as wheelchair movement status and obstacle distances to the mobile dashboard without significant delay. This feature allows caregivers to monitor the user’s condition and surroundings remotely, improving overall safety and support. The wireless communication via ESP32 was stable under normal network conditions, although performance may vary with internet connectivity.

The use of RTOS in the ESP32 ensured efficient multitasking and smooth system operation. Different tasks such as voice processing, sensor monitoring, and IoT communication were executed simultaneously without conflicts. This resulted in faster response time and better system coordination. The priority-based task scheduling ensured that critical operations like emergency stop and obstacle detection were handled promptly.

Overall, the results indicate that the proposed system is effective, reliable, and suitable for real-time applications. It successfully combines control, safety, and monitoring features into a single platform. While there are minor limitations related to sensor accuracy and environmental conditions, the system performs well under typical usage scenarios. The project demonstrates the potential of integrating IoT and intelligent control techniques in assistive devices to enhance user independence, safety, and quality of life.

CHAPTER 8: ADVANTAGES AND DISADVANTAGES

8.1 Advantages

- Provides hands-free control using voice and gesture
- Suitable for physically disabled and elderly users
- Multiple control methods increase flexibility and reliability
- Obstacle detection system prevents collisions
- Buzzer alert improves safety awareness
- Emergency voice command ensures immediate stopping
- IoT integration (Blynk) enables real-time monitoring
- Allows remote supervision by caregivers
- Cost-effective compared to advanced commercial wheelchairs
- Uses easily available components (ESP32, sensors)
- RTOS-based system ensures fast and efficient multitasking
- User-friendly and easy to operate
- Modular design allows future upgrades

8.2 Disadvantages

- Voice recognition may be affected by noise interference
- Works only with predefined voice commands
- Gesture control may have accuracy issues due to unwanted movements
- Requires user training for gesture usage
- Ultrasonic sensors have limited range and accuracy
- May not detect small or soft obstacles properly
- Battery dependency requires regular charging
- Continuous Wi-Fi use may drain battery faster
- IoT features depend on stable internet connection
- Lacks advanced AI-based navigation
- Performance may reduce in complex real-world environments

8.3 Applications

- Used by physically disabled individuals for mobility
- Helpful for elderly people with limited movement
- Used in hospitals for patient transportation
- Applied in rehabilitation centers
- Suitable for home care environments
- Enables remote patient monitoring using IoT
- Used in smart healthcare systems
- Useful in assistive technology research
- Can be extended to robotics and automation systems
- Supports smart city accessibility solutions

CHAPTER 10: FUTURE WORK

The proposed system can be further enhanced by integrating advanced technologies to improve functionality and user experience. Features such as GPS-based tracking, AI-powered voice recognition, and autonomous navigation can make the wheelchair more intelligent and efficient. Incorporating advanced sensors and health monitoring systems can improve safety and usability. Additionally, better power management and cloud integration can enhance performance, making the system more suitable for real-world applications.

- Integration of GPS for real-time location tracking
- Implementation of LiDAR or camera-based obstacle detection
- AI-based voice recognition for improved accuracy
- Development of a dedicated mobile application for control
- Autonomous navigation using path planning algorithms
- Addition of health monitoring sensors (heart rate, temperature)
- Improved battery life and power management system
- Cloud data storage for analysis and history tracki

CHAPTER 11: SOURCE CODE

```
// ==========================================
// BLYNK & WIFI CONFIGURATION (Must be at top)
// ==========================================
#define BLYNK_TEMPLATE_ID "TMPL3L5cPvFFM"       // Replace with your ID
#define BLYNK_TEMPLATE_NAME "wheelchair" // Replace with your Name
#define BLYNK_AUTH_TOKEN "QyKslJEOJOrffybXm_toW1sAiOW_LxWj"      //
Replace with your Token

#include <WiFi.h>
#include <WiFiClient.h>
#include <BlynkSimpleEsp32.h>
#include <Wire.h>
#include <MPU6050.h>

char ssid[] = "Vivo";
char pass[] = "01234567";

// ==========================================
// PIN DEFINITIONS
// ==========================================
// Motors (L293D)
#define IN1 27
#define IN2 26
#define IN3 25
#define IN4 33

#define buzzer 32
// Ultrasonic Front
```

```
#define TRIG_FRONT 5
#define ECHO_FRONT 18
// Ultrasonic Back
#define TRIG_BACK 19
#define ECHO_BACK 23

// Voice Module (Serial2 Default)
#define RX2_PIN 16
#define TX2_PIN 17

// ==========================================
// CONSTANTS & GLOBALS
// ==========================================
const int DISTANCE_THRESHOLD = 20; // cm

// System States
enum Direction { STOPPED, FORWARD, BACKWARD, LEFT, RIGHT };
volatile Direction currentDir = STOPPED;
volatile bool voiceActive = false; // Arbitration flag: Voice > Gestures

// Notification Cooldown
unsigned long lastNotifyTime = 0;

// MPU6050
MPU6050 mpu;
int16_t ax, ay, az;
int16_t ax_off, ay_off, az_off;

// Front Sensor
volatile long startFront, endFront;
```

```
volatile bool echoFlagFront = false;
volatile bool obstacleFront = false;

// Back Sensor
volatile long startBack, endBack;
volatile bool echoFlagBack = false;
volatile bool obstacleBack = false;

// RTOS Task Handles
TaskHandle_t TaskVoice_Handle;
TaskHandle_t TaskObstacle_Handle;
TaskHandle_t TaskGesture_Handle;
TaskHandle_t TaskBlynk_Handle;

// ==========================================
// HARDWARE INTERRUPTS (ISRs)
// ==========================================
void IRAM_ATTR echoFrontISR() {
 if (digitalRead(ECHO_FRONT) == HIGH) startFront = micros();
 else { endFront = micros(); echoFlagFront = true; }
}

void IRAM_ATTR echoBackISR() {
if (digitalRead(ECHO_BACK) == HIGH) startBack = micros(); else
 { endBack = micros(); echoFlagBack = true; }
}

// Function Prototypes
void calibrateMPU();
void moveForward();
void moveBackward();
```

```
void turnLeft();
void turnRight();
void stopMotors();

// ==========================================
// SETUP
// ==========================================
void setup() {
Serial.begin(115200);
 Serial2.begin(9600, SERIAL_8N1, RX2_PIN, TX2_PIN); // VC-02
// Pin Modes
 pinMode(IN1, OUTPUT); pinMode(IN2, OUTPUT);
 pinMode(IN3, OUTPUT); pinMode(IN4, OUTPUT);
 pinMode(TRIG_FRONT, OUTPUT); pinMode(ECHO_FRONT, INPUT);
 pinMode(TRIG_BACK, OUTPUT);  pinMode(ECHO_BACK, INPUT);
 pinMode(buzzer, OUTPUT);
 // Attach Interrupts
 attachInterrupt(digitalPinToInterrupt(ECHO_FRONT), echoFrontISR, CHANGE);
 attachInterrupt(digitalPinToInterrupt(ECHO_BACK), echoBackISR, CHANGE);

 digitalWrite(buzzer, LOW);
// Initialize I2C & MPU6050
 Wire.begin(21, 22);
 mpu.initialize();
 if (!mpu.testConnection()) { Serial.println("MPU6050
  failed. Check wiring."); while (1) { vTaskDelay(10);
  }
  }
  calibrateMPU();
```

```
// Ensure motors are off at boot
stopMotors();
Serial.println("System Ready - Initializing WiFi and RTOS Tasks...");
WiFi.begin(ssid, pass);
while(WiFi.status()!=WL_CONNECTED)
{
 Serial.print("..");
}
Serial.println("Connected");

// Initialize Blynk (Blocking until connected initially, or you can use WiFi.begin for custom non-blocking)
Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
Serial.println("Blynk Connected");

// --- Task Creation ---
// Core 1, Priority 4 (HIGHEST): Emergency Stop & Voice Control
xTaskCreatePinnedToCore(TaskVoice, "VoiceTask", 4096, NULL, 4, &TaskVoice_Handle, 1);

// Core 0, Priority 3: Dual Obstacle Detection
xTaskCreatePinnedToCore(TaskObstacle, "ObstacleTask", 4096, NULL, 3, &TaskObstacle_Handle, 0);

// Core 0, Priority 2: Blynk Background Task
xTaskCreatePinnedToCore(TaskBlynk, "BlynkTask", 8192, NULL, 2, &TaskBlynk_Handle, 0);

// Core 1, Priority 1 (LOWEST): Continuous Gesture Polling
xTaskCreatePinnedToCore(TaskGesture, "GestureTask", 4096, NULL, 1, &TaskGesture_Handle, 1);
}
```

```
void loop() {
vTaskDelete(NULL); // Loop is not needed in RTOS
}

// ==========================================
// RTOS TASKS
// ==========================================

// Task 1: Blynk Connection Manager (Core 0)
void TaskBlynk(void *pvParameters) {
for (;;) {
if (WiFi.status() == WL_CONNECTED) { if
(!Blynk.connected()) {
Blynk.connect(); // Attempt to reconnect if lost
Serial.println("Connecting. .....");
} else {
Blynk.run();// Process incoming/outgoing Blynk data
}
}
// Yield to the Obstacle task
vTaskDelay(pdMS_TO_TICKS(100));
}
}

// Task 2: Voice Commands (Highest Priority - Core 1)
void TaskVoice(void *pvParameters) {
unsigned int receivedvalue = 0;
for (;;) {
while (Serial2.available()) { byte
high = Serial2.read();
```

```
byte low = Serial2.read();

receivedvalue = (high << 8) | low;


// PRIORITY 1: Emergency Stop

if (receivedvalue == 0xAA11 || receivedvalue == 0xBB22) {

Serial.println("VOICE COMMAND: STOP!"); stopMotors();

if(receivedvalue == 0xBB22){

// Push notification to Blynk (with 10-sec cooldown) if

(millis() - lastNotifyTime > 10000) {

Blynk.logEvent("emergency");

lastNotifyTime = millis();

}

}

voiceActive = false; // Release control back to Gestures

// Flush buffer

while(Serial2.available()) Serial2.read();

}

// Normal Voice Commands

 else if (receivedvalue == 0xA189) { voiceActive = true; Serial.println("VOICE
COMMAND: Forward");moveForward();   }

else if (receivedvalue == 0xA191) { voiceActive = true;Serial.println("VOICE
COMMAND: Backward"); moveBackward(); }

else if (receivedvalue == 0xA020) { voiceActive = true; Serial.println("VOICE
COMMAND: Left");turnLeft();   }

else if (receivedvalue == 0xA030) { voiceActive = true; Serial.println("VOICE
COMMAND: Right");turnRight();  }

}

vTaskDelay(pdMS_TO_TICKS(5));

}

}


// Task 3: Obstacle Detection (Core 0)
```

```
void   TaskObstacle(void   *pvParameters)
{ float distFront = 100.0, distBack = 100.0;
unsigned long lastGaugeUpdate =      0;
for (;;) {
// 1. Ping Front Sensor
digitalWrite(TRIG_FRONT, LOW); delayMicroseconds(2); digitalWrite(TRIG_FRONT,
HIGH); delayMicroseconds(10); digitalWrite(TRIG_FRONT, LOW);

vTaskDelay(pdMS_TO_TICKS(30));

if (echoFlagFront) {
distFront = (endFront - startFront) * 0.034 / 2;
echoFlagFront = false;
obstacleFront = (distFront > 0 && distFront < DISTANCE_THRESHOLD);
}

// 2. Ping Back Sensor
digitalWrite(TRIG_BACK, LOW); delayMicroseconds(2);
digitalWrite(TRIG_BACK, HIGH); delayMicroseconds(10);
digitalWrite(TRIG_BACK, LOW);

vTaskDelay(pdMS_TO_TICKS(30));

if (echoFlagBack) {
distBack = (endBack - startBack) * 0.034 / 2;
echoFlagBack = false;
obstacleBack = (distBack > 0 && distBack < DISTANCE_THRESHOLD);
}
// Send data to Blynk every 500ms
```

```
// Send data to Blynk every 500ms
if (millis() - lastGaugeUpdate > 500) {
// V1 for Front Distance, V2 for Back Distance
Blynk.virtualWrite(V1, distFront);
Blynk.virtualWrite(V2, distBack);

// Determine string based on current direction String
dirString = "Stopped";
if (currentDir == FORWARD) dirString = "Forward";
else if (currentDir == BACKWARD) dirString = "Backward"; else
if (currentDir == LEFT) dirString = "Left";
else if (currentDir == RIGHT) dirString = "Right";

// Send Current Direction to V0
Blynk.virtualWrite(V0, dirString);

lastGaugeUpdate = millis(); // Reset the timer
}
// --- BUZZER LOGIC ---
if (obstacleFront || obstacleBack) {
digitalWrite(buzzer, HIGH); // Sound buzzer if obstacle detected
} else {
digitalWrite(buzzer, LOW);  // Turn off if path is clear
}
// 3. Dynamic Stop Logic
if (obstacleFront && (currentDir == FORWARD || currentDir == LEFT || currentDir ==
RIGHT)) {
Serial.println("ALERT: FRONT OBSTACLE! Stopping Motion"); stopMotors();
}
if (obstacleBack && currentDir == BACKWARD) { Serial.println("ALERT:
REAR OBSTACLE! Stopping Motion");
```

```
stopMotors();
}
}
}

// Task 4: Gestures (Core 1)
void TaskGesture(void *pvParameters) {
int ax_map, ay_map;
for (;;) {
// Only process if Voice hasn't taken over
if(!voiceActive) {
mpu.getAcceleration(&ax, &ay, &az);  ax -
= ax_off; ay -= ay_off;
ax_map = map(constrain(ax, -16384, 16384), -16384, 16384, 0, 1000);
ay_map = map(constrain(ay, -16384, 16384), -16384, 16384, 0, 1000);

if (ax_map > 650 && ax_map < 850) {
if (!obstacleFront){ Serial.println("GESTURE COMMAND: Forward");moveForward();}
} else if (ax_map < 150 && ax_map < 350) {
if (!obstacleBack){Serial.println("GESTURE COMMAND: Backward");
moveBackward();}
} else if (ay_map > 650 && ay_map < 850) {
if (!obstacleFront) {Serial.println("GESTURE COMMAND: LEFT");turnLeft();}
} else if (ay_map < 150 && ay_map < 350) {
if (!obstacleFront) {Serial.println("GESTURE COMMAND: Right");turnRight();}
} else {
Serial.println("GESTURE COMMAND: STOP");
stopMotors();
}
}
```

```
vTaskDelay(pdMS_TO_TICKS(100)); // Poll 10x a second
}
}
// ==========================================
// MOTOR CONTROL & INTERLOCKS
// ==========================================
void moveForward() {
if (obstacleFront) return;
if (currentDir != FORWARD) {
Serial.println("MOTOR Action: Forward");
currentDir = FORWARD;
digitalWrite(IN1, HIGH); digitalWrite(IN2, LOW);
digitalWrite(IN3, HIGH); digitalWrite(IN4, LOW);
}
}
void moveBackward() {
if (obstacleBack)
return;
if (currentDir != BACKWARD) { Serial.println("MOTOR
Action: Backward"); currentDir = BACKWARD;
digitalWrite(IN1, LOW);  digitalWrite(IN2, HIGH);
digitalWrite(IN3, LOW);  digitalWrite(IN4, HIGH);
}
}
void turnLeft() {
if (obstacleFront) return;
if (currentDir != LEFT) {
Serial.println("MOTOR Action: Left");
currentDir = LEFT;
digitalWrite(IN1, LOW);  digitalWrite(IN2, HIGH);
```

```
}
}
void turnRight() {
if (obstacleFront) return;
if (currentDir != RIGHT) {
Serial.println("MOTOR Action: Right");
currentDir = RIGHT;
digitalWrite(IN1, HIGH); digitalWrite(IN2, LOW);
digitalWrite(IN3, LOW);  digitalWrite(IN4, HIGH);
}
}
void stopMotors() {
if (currentDir != STOPPED) {
currentDir = STOPPED;
Serial.println("MOTOR Action: !!!STOP!!!");
digitalWrite(IN1, LOW); digitalWrite(IN2, LOW);
digitalWrite(IN3, LOW); digitalWrite(IN4, LOW);
}
}
// ==========================================
// CALIBRATION
// ==========================================
void calibrateMPU() {
long ax_s = 0, ay_s = 0, az_s = 0;
for (int i = 0; i < 100; i++) {
mpu.getAcceleration(&ax, &ay, &az);
ax_s += ax; ay_s += ay; az_s += az;
vTaskDelay(pdMS_TO_TICKS(2));
}
ax_off = ax_s / 100; ay_off = ay_s / 100; az_off = (az_s / 100) - 16384;
}
```

CHAPTER 9: CONCLUSION

The proposed gesture and voice controlled smart wheelchair with IoT monitoring and obstacle detection using ESP32 demonstrates an effective solution for improving mobility assistance for physically challenged and elderly individuals. By integrating both gesture and offline voice control, the system provides a flexible and user-friendly interface that reduces the need for physical effort and external assistance. The use of the MPU6050 sensor and voice recognition module ensures reliable input methods, allowing users to operate the wheelchair conveniently in different situations. The system places strong emphasis on safety through continuous obstacle detection using ultrasonic sensors. The automatic stopping mechanism and buzzer alert significantly reduce the risk of collisions, ensuring secure navigation even in confined environments. The implementation of Free RTOS enables efficient multitasking, allowing simultaneous execution of control, sensing, and communication tasks without delay. In addition, the integration of IoT through the Blynk platform provides real-time monitoring of wheelchair status and obstacle distances, enhancing user awareness and remote supervision. The system is designed to be cost-effective, energy-efficient, and easy to use, making it suitable for real-world applications. Overall, the project successfully combines embedded systems and IoT technologies to create a smart, reliable, and safe wheelchair solution that enhances independence, comfort, and quality of life for users.

CHAPTER 12: REFERENCES

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